

WHAT IS _____

Curated by 64/1



FOUNDATION
WE CARE FOR THE FUTURE

WHAT IS ____?

CURATED BY 64/1

FEATURING
HARSHIT
RAGHAVA KK
KARTHIK KALYANARAMAN

POTENTIAL PARTNERS



WHAT IS BLACK ?

LOVE
WOMAN
SPACE
OBJECT
FAITH

15th January - 15th March 2020

Presented by **RMZF**
Curated by **64/1**

Dr. Karthik Kalyanaraman & Raghava KK

THE EXPERIENCE

More than just a show:

CONCEPT

Do universal concepts like womanness, color, laws of physics 'exist'? This has been a perennial question in Indian and Western philosophy. We explore this question with, not humans, but the new 'thinking thing' in our lives: AI

Important to note: will include themes from the United Nations Global Compact + involve Traditional Artisans and artists (artisans - metal work, pottery, ceramics, etc)

EXHIBITION

6 works

Each work = art installation + Data installation + Wall text + how the work was created

EDUCATION

Collaboration with Sristi School of Design, Bangalore

Two public talks

OUTREACH

Art press and international coverage from Business press, Art press, Lifestyle press and Tech press.

ARCHIVE

Brochure for show written by 64/1 and art critic

AI basics 101 instruction guide

A NEW ART PRACTICE

The show brings together pioneering Indian AI artists, artisans and curators who have adopted a radically new aesthetic practice: their art process is created by practice of artificially intelligent(AI) algorithm working closely with traditional Indian artisans and contemporary artists and curators.

We pioneered the first global gallery exhibition of AI art. However we want to take AI art to the next level where it's not just about AI but has considerable conceptual depth as well. Aside from the social consequences of AI, we aim to explore the philosophical challenges for the future of aesthetics posed by the very existence of AI art.

What is AI art?

The kind of AI algorithm used by the artists, is called a neural network: its design was meant to mimic the activity of biological neurons. To teach a neural network a visual language, the artist does not provide it with any rule or principle; she only exposes it to a large number of examples (called training set) and allows it to draw its conclusions. After this 'training', the AI is then ready to start creating the final work.

The training is by no means a simple process. Aside from the role of creating the conceptual basis of the artwork, it is here that the human artist's novel practice is situated. This is the art of the statistician; the art of the dataset, of choosing the examples one uses for 'training'. There are several issues here: how to curate/create the training set, how to impose visual constraints on it, how to adjust how quickly the algorithm learns. And all this is done with a more traditional aesthetic eye on the quality of the work the AI creates. In any case, the putative gap between the 'creative' practices of the artist and the supposedly logical practice of the scientist is problematized.

IMPACT

THE DIRECT IMPACT FOR THE BUSINESS WORLD AND THE ARTWORLD

- Our future is **AI**. We need to confront this. If the machine becomes expert even at creating art, what is left for human artists to do? This is a question and a problem of great consequence for the artists of the future. This show will provide at least one way forward: become cyborg.
- We believe that art trends can be started in India and rooted in Indian aesthetics and we do not need to continue copying Western trends. This exhibition is conceived at something at the cutting edge of **global** art.
- We live in a time dominated by divisive politics. This show disturbs settled **identity** categories like caste, gender, etc. with the goal of creating larger awareness of issues that face all of humanity like the rise of AI
- We engage traditional **artisans** with this art of the future, so we can see AI as a tool to enrich traditional arts rather than something that will displace them

TALKS

Two talk - 2 hours each + Gallery walkthrough + Gallery manager training

Talk 1

Title: From the past towards a future

A conversation between artisans, artists and technologists to see how they can co-exist/co-create - how the past and the future create a magnificent present

Speakers: Karthik Kalyanaraman, Raghava KK, Amrita Varma, A ceramicist, Padmanabhan, A mestri (master craftsman) and Harshit Agarwal

Format - Short introduction followed by a moderated conversation - 1 hour each - two sessions.

1. Conversations with artists
2. Conversations with artisans
3. Public q&a
4. Final thoughts by the curators
5. Walkthrough of exhibit

Audience: Gallery show opening invitees.

Talk 2

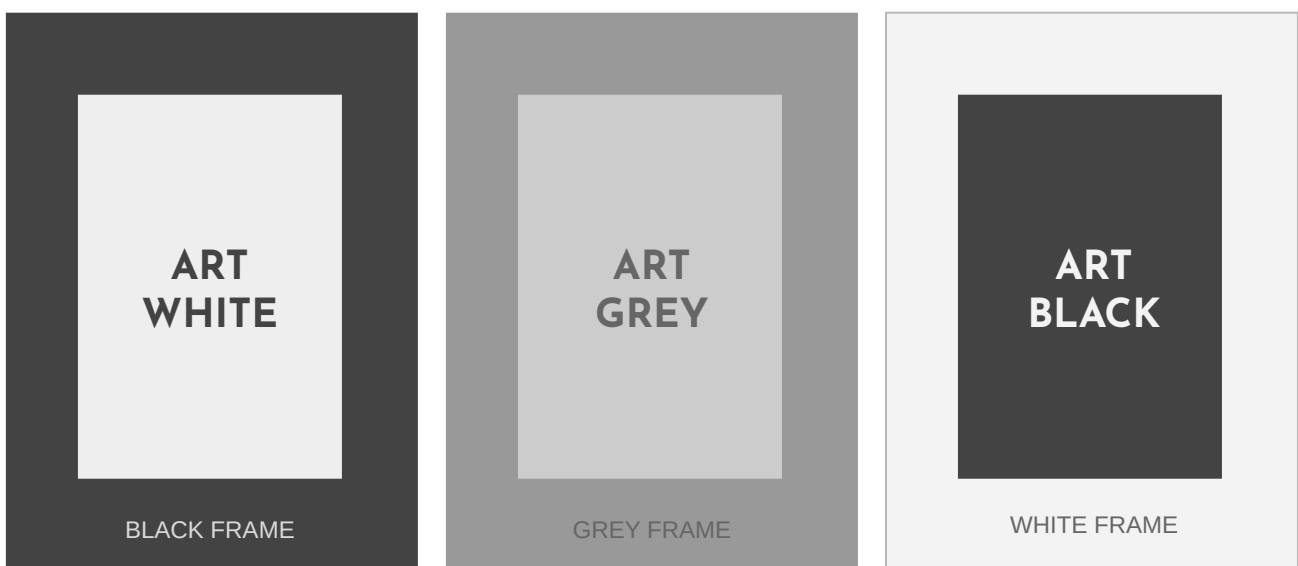
Title: 'Yagna' @ RMZF

A new experimental talk format. Details to follow.

Audience: Both RMZF and 64/1 will send out invitation to the following audience
Art collectors, artists, students of art, NGOs that support artisans, RMZ tenants and team.

SAMPLE ARTWORK #1

WHERE CAN BLACK BE FOUND?



How does an AI imagine the color black? White? It turns out even for machines all imaginations of concepts involve their opposites! We explore what a machine's understanding of color is.

Data Set of 10,000 curated images from the internet of images tagged #black
Hand-painted by a billboard painter

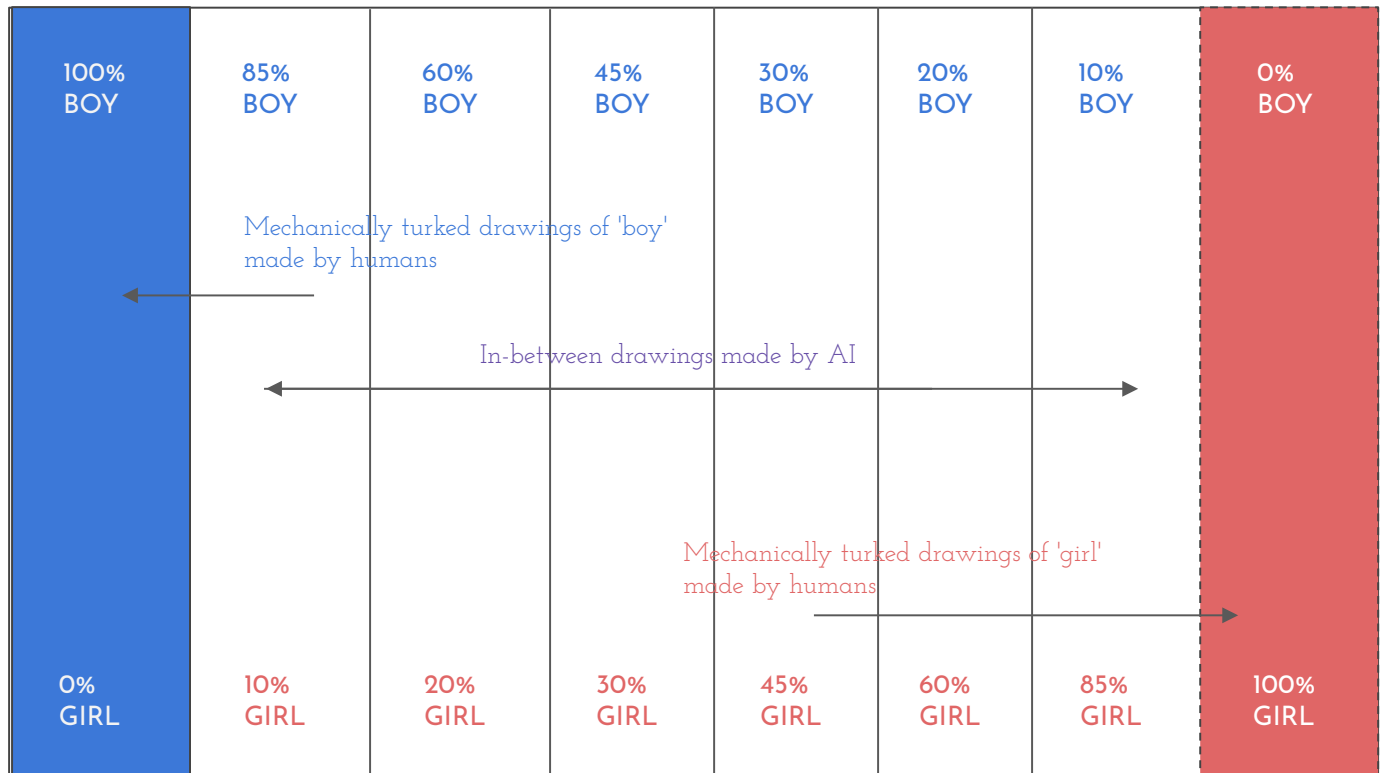
Frame:

Material determined by color material AI trained on a Data Set of 10,000
curated images from the internet of images of materials that indicate #black
Handmade by local framer



SAMPLE ARTWORK #2

MACHINE DREAMS OF WOMAN



Once an AI Algorithm has learned about how humans draw abstract representation of male and female humans, it can surprisingly also produce original drawings of humans in between indicating a very non-binary mode of conceptualization of gender.

Frame : Industrial steel frame

Inside frame: Steel mesh with hand painted tiles

Inside steel mesh: Small painted ceramic tiles

Size of installation: (TBD approx 8ft X 15 feet)

In Between Drawings Created by an AI algorithm

Data Set 10,000 Mechanically turked images of Boy and Girl
outsourced to Amazon Mechanical turk
Printed

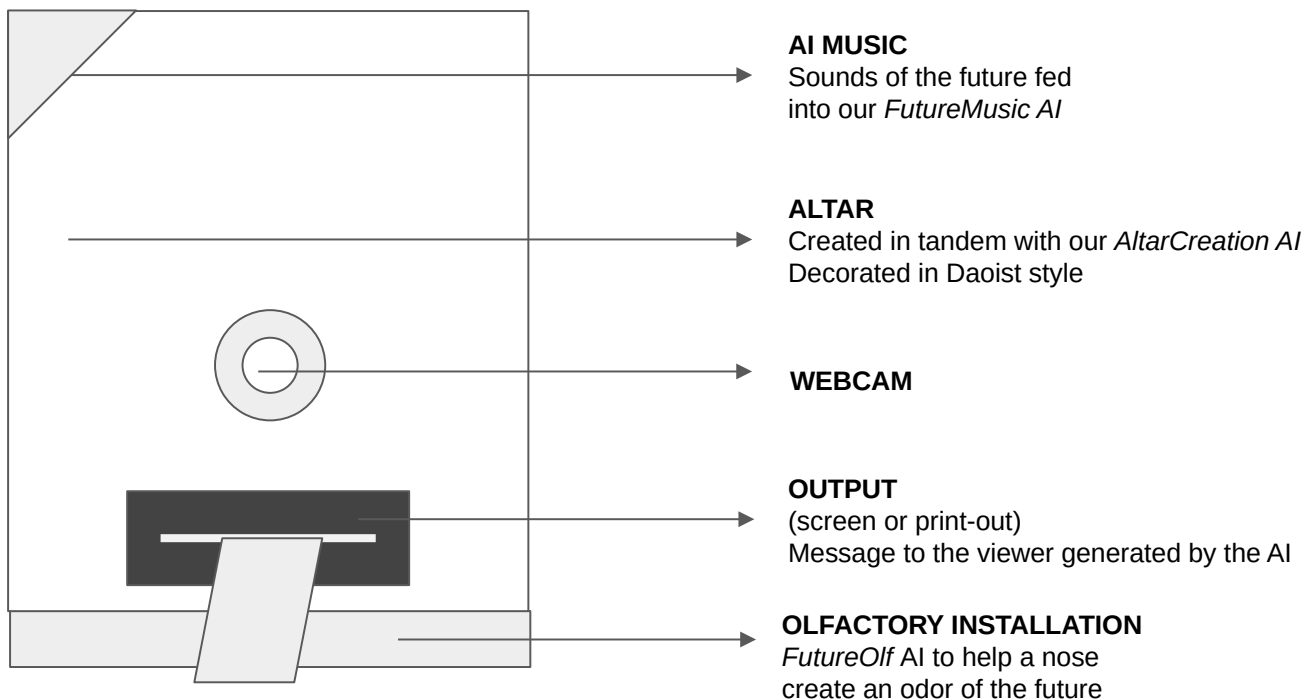
Tiles arrangement:

Tiles fade from blue to pink as the percentage of boy/girl



SAMPLE ARTWORK #3

ALTAR TO A FUTURE

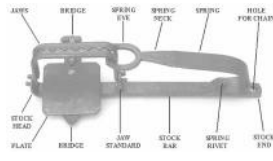


What are the forces that will dominate our future? In other words what are the effective god-like entities of the future? Is it climate change? AI itself ? We explore this question with a multi-sensory interactive installation where each element (sound, interaction and smell) are created in collaboration with different AI algorithms. The altar will interact by text with each visitor.

Dataset: AI algorithm that learns concepts of smell associated with the future; dataset of sound files using will an AI will create music of the future and images of the future.

SAMPLE ARTWORK #4

ONCE THE ANIMALS DEPART



OBJECT 1
A large replica of an
elephant snare

10,000 images feed the
ReImagineObject AI



OBJECT 2
The object studied and
reimagined by AI, created
by craftsman

10,000 images feed the
ReImagineObject AI



OBJECT 3
The object 2 studied and
reimagined by AI, created
by craftsman

When AI looks at a 3D object, we reconstruct it using 2D perspectives, just like a human would. This the fascinating thing is that even a particular object is an abstraction.! When we think of our future two alarming t hints about it are the loss of animal species and the rise of potentially super intelligent AI. How would an AI imagine and reimagine an animal snare? And what would an abstraction of an abstraction look like?

Dataset:

Thousands of photographs of the objects taken under different lighting conditions, distances and perspectives. Photographers come from all sections of society.

THE CURATORS

64/1

64/1 - A curatorial collective focused on art for the Post-Human age

64/1 is an art curation and research collective focused on building public understanding of how artists and artificial intelligence can come together to create art for the post-human age. They have written the first academic document coming out of the Indian art world addressing the ability of contemporary art to help create an experience of a dynamic human-machine relationship. 64/1 was founded by Raghava KK and Karthik Kalyanaraman out of Bangalore, India. Raghava KK is a multidisciplinary artist, TED speaker and TED curator. Karthik Kalyanaraman is a former professor of econometrics at University College London and University of Maryland, College Park.

THE ARTISTS



Dr. Karthik Kalyanaraman co-founded 64/1 along with his brother Raghava KK. He started by studying Art History but ended up completing his PhD in Econometrics from Harvard University, and was a professor of the same at University College London and University of Maryland, College Park. Dr. Kalyanaraman is also a curator, creator of AI art, and an experience designer.



Raghava KK was named by CNN as one of the 10 most remarkable people in 2010. He is a multidisciplinary artist and entrepreneur working at the intersection of art, science, technology, education and entrepreneurship.

Raghava is a National Geographic Explorer and a multiple-time TED speaker.

He along with his brother Dr. Karthik Kalyanaraman co-founded 64/1, a research collective focused on art for the post human age. Their debut curatorial project Gradient Descent recognized as the first AI Art exhibition in India, and perhaps one of the first serious ones at a gallery, attracted pioneers from 9 countries. Their paper 'Transcendence for the Siri-ocene' was published at India's leading aesthetics journal Critical Collective.

Raghava and his brother also co-authored the acclaimed board game Shasn.

Popular Netflix show "The Creative Indians" dedicated an episode on his art and life.



Harshit Agrawal is a new media artist and human computer interaction (HCI) researcher. Through his artwork, he wants to create experiences for people to explore and express with seemingly distant technologies like artificial intelligence/ machine learning, drones, digital fabrication, sensors, augmented reality and in the process invite people to reflect upon and re-evaluate their relationship with technology.

He believes we live within a complex sphere of technical processes, and seldom do our experiences make us stop and come face to face with them, sit down and have a conversation with them, become comfortable with them to a level that we start reflecting on how they are truly affecting us.

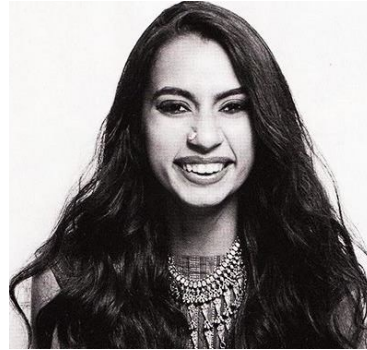
As an artist, he wants to raise questions around such influences that technology has on us.

THE ARTISTS



SiriDevi Khandavalli works across painting, performance, video, and installation to create visual art that is rooted in traditional Indian techniques and yet transcends cultural specificity to hint at larger commentaries about the materialistic nature of the present. She is best known for her cheeky Chola-like bronzes.

Other notable works include *Eat* (2009), a looped video critiquing mindless consumption, and *Lucky Lakshmi Dollar Bill* (2010), a conceptual work created from US currency that emphasizes the multiple connotations of money across cultures. Seen as a whole, Khandavalli's diverse body of work is centered on the notion of desire and seeks to answer the question of what is it that moves people and the world.



Shilo Shiv Suleman is a multi award winning artist, INK fellow and TED speaker with a focus on the intersection of magical realism, art for social change and technology. In most recent years, she's been engaging with biofeedback technology, and the interaction between the body and art.

She has created large scale installations that beat with your heart, apps that react to your brainwaves and sculptures that glow with your breath. She's been featured in a host of documentaries, and been recipient of several international grants.

THE COLLABORATORS

Artisans

Master Framer designer
Billboard artist/painter
Mestry
Ceramicist
Anonymous Mechanical Turks from Amazon Web Services
A Nose (master parfumeur)
Metal craftsman

Others

Graphic designer
Archivist
Hardware technician

AI

Each artwork will have dedicated expert AIs built and trained for the purpose

TIMELINE

September	October	November	December	January	February	March	April
Sign Contract	Concept Design	Collateral + Website	Conversation Starter	Launch	Pilot		Creative Lab - Path Forward
	Production		PR + Social Media	Exhibit			

BUDGET + DELIVERABLES

BUDGET SECTION	DETAILS	AMOUNT
6 AI ART WORKS (@ avg. 4.5L production cost per work)	RESEARCH + PROTOTYPES + DATA INSTALLATION	27,00,000
	ART SUPPLIES	
	ARTISTS FEE + ARTISANS FEE	
	ARTISTS TRAVEL	
	WALL TEXT	
CURATION FEE	64/1 - 4 TEAM MEMBERS ENGAGED FOR 4 MONTHS	5,00,000
DOCUMENTATION	BROCHURE DESIGN, CONTENT, PHOTOGRAPHY AND VIDEOGRAPHY	2,60,000
TECH RIDERS		40,000
TWO PUBLIC TALKS	GOODWILL GESTURE	0
TOTAL		29,66,600
GST (18%)	ROUNDED OFF	5,33,400
TOTAL INCL. GST		3500000

WHAT RMZF GIVES

Exhibition production and curation sponsorship (29.66L)
 PR
 Shristi connection and relationship
 Permission for us to bring in other sponsors
 Permission to both 64/1 and RMZF to sell works. 15% of sale value minus tax to the foundation towards the Creative Lab if sale is made during the show.
 A point of contact for logistics and permits etc.

WHAT RMZF GETS

Show
 2 talks
 1 artwork of your choice - to be decided before the show opens. First preference for choice will go to RMZF + show @ Rs.

Payment Timeline:

50% of the total amount will be paid as an advance on **October 1**.
 The next **40%** by **January 15**
 The final **10%** by **March 20**
 The advance amount will not be refunded in case the event is cancelled.
 If the event date is moved, our involvement is subject to our availability.
 5% additional fee if payment is delayed by 30 days.

THANK YOU

We are tremendously excited at how our ideals are perfectly aligned

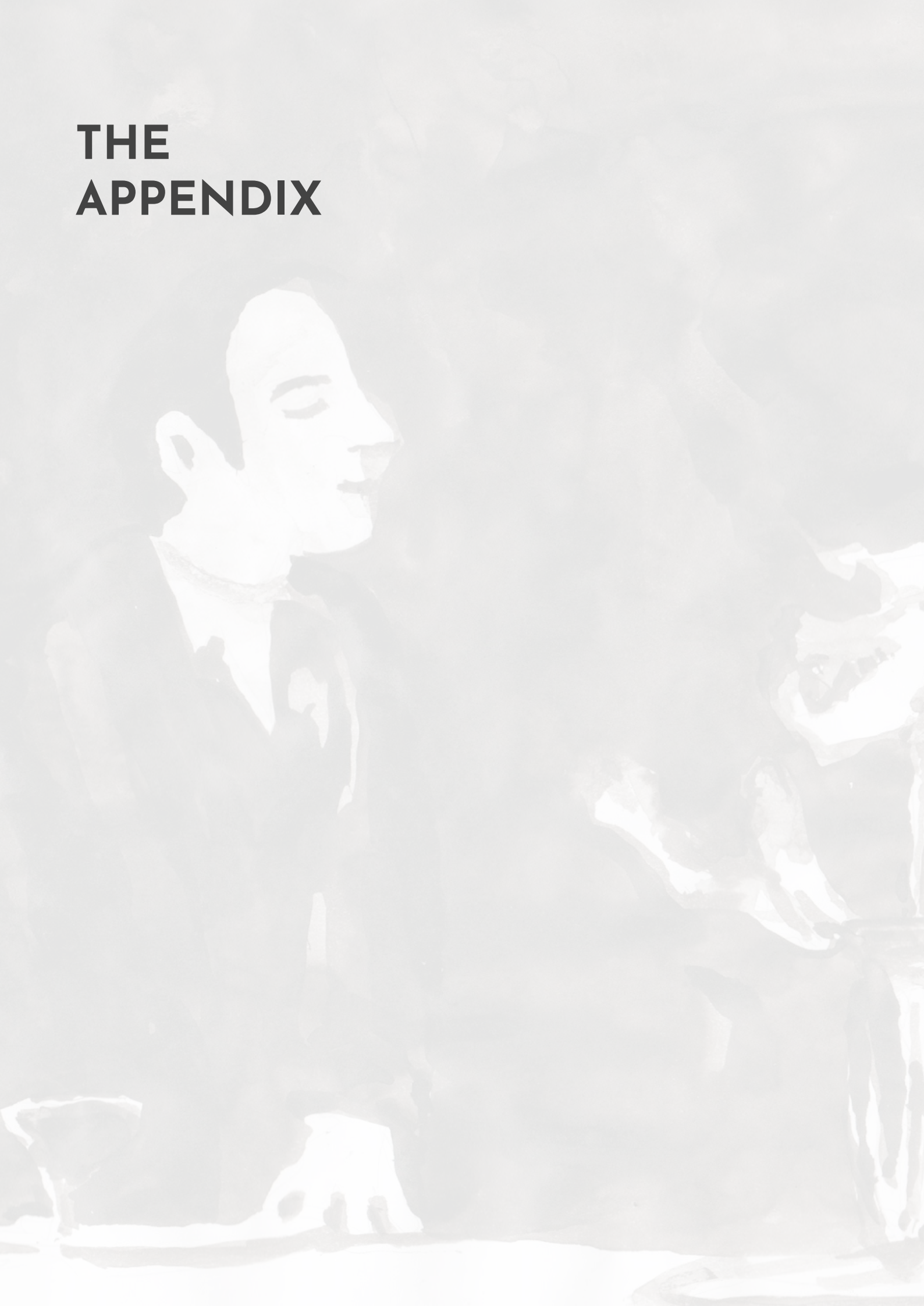
We look forward to creating the vision of an exciting future with you.

64/1 FOR



FOUNDATION
WE CARE FOR THE FUTURE

THE APPENDIX



PRESS FROM 64/I's AI ART SHOW (2018)

The New York Times

Up for Bid, AI Art Signed 'Algorithm'



"Edmond de Belamy, from La Famille de Belamy," a print on canvas produced by the French collective Obvious, working with artificial intelligence, will be auctioned Thursday at Christie's. Credit: Christie's

By Gabe Cohn
Oct. 22, 2018

With their eyes trained on a mild frame containing a smeared, half-formed image of a distinguished gentleman, a small group of potential bidders gathered Friday night over cocktails at Christie's New York and heard the pitch: here was the first portrait generated by an algorithm to come up for auction. The portrait, produced by artificial intelligence, hung on the wall opposite an Andy Warhol print and just to the right of a bronze work by Roy Lichtenstein.

There were some smiles and at least one frown. Two people snapped cellphone pictures of the work, which looked as though someone had taken a sponge to a 17th-century oil portrait. The arrival of what some call the infant stages of the next great art movement at one of the world's leading auction houses was greeted nonchalantly, with a nod of understanding and a sip of mezcal margarita.



"Perception Engines," by Tom White, another AI artist, was shown at the New Delhi gallery Nature Morte in 2018. Credit: Nature Morte, New Delhi

The opportunity to capitalize on the buzz surrounding artificial intelligence fits in with Christie's effort to market itself in new ways.

In July it held its [first Art & Tech summit](#), focused on the theme of blockchain, the technology behind cryptocurrencies like Bitcoin. (Obvious themselves have sold art on [SuperRare](#), an online marketplace that allows users to buy and sell digital artwork using cryptocurrency.)

Brick-and-mortar auction houses have been slow to accept art that explores new technologies. To be sure, Sotheby's, Christie's and Phillips offer plenty of digitally generated works by blue-chip artists like Andreas Gursky and Christopher Wool. But the sort of digital video and virtual reality pieces that are featured routinely at contemporary art biennials rarely, if ever, appear at auction houses.

Artists in the field are hoping that works sold at auctions like Christie's will increasingly enter private collections. That appears to be happening more often. Earlier this year the New Delhi art gallery Nature Morte hosted what was billed as [India's first exhibition](#) of work made entirely by AI, including works by Mr. Klingenstein, Tom White, Memo Akten, Jake Elwes and Anna Rieger. In December, Dr. Elasmund of Rutgers will exhibit AI art for sale at [SCOPE Miami Beach](#), a satellite fair during the time of Art Basel Miami Beach.

Who knows what art historians of the future will say about these works and the impact of AI as a tool for artists. But Dr. Bivansgarnier at Columbia offered a gut response to the Obvious portrait. "It's just so strange," she said. *Correction: October 23, 2018*

An earlier version of this article misstated where AI work will be exhibited for sale in Miami. It will be shown at Scope Miami Beach, not Art Basel Miami Beach.

<https://www.nytimes.com/2018/10/22/arts/design/christies-art-artificial-intelligence-obvious.html>

<https://www.nytimes.com/2018/10/22/arts/design/christies-art-artificial-intelligence-obvious.html>

The New York Times

Heading : Up for Bid, AI Art Signed 'Algorithm'

Art Market

When Art Created by Artificial Intelligence Sells, Who Gets Paid?

Sam Gaskin

Sep 17, 2018 11:45 am

Christie's will auction off an artificial intelligence (AI) artwork for the first time this October, hard on the heels of a pioneering all-AI art exhibition held at New Delhi gallery Nature Morte. While the market is eager to move the work, the field raises questions about ownership, obsolescence, and the art world jobs that algorithms can't do.

What is AI art?



https://www.artsy.net/article/artsy-editorial-art-created-artificial-intelligence-sells-paid?utm_medium=email&utm_source=14477918_newsletter_editorial-daily-09-17-18&utm_campaign=editorial&utm_content=st-P

Artsy

Headline:

When Art Created by Artificial Intelligence Sells, Who Gets Paid?

Forbes

Artificial Intelligence Art: Parenting a new genre

The first-ever show of art created by Artificial Intelligence just opened at the Nature Morte Gallery in New Delhi. Presented by Aparajita Jain, Gradient Descent brings together seven pioneering international artists who have adopted a radically new aesthetic practice: Their final artwork is actually created by an AI algorithm. The show is curated by Raghava KK and Karthik Kalyanaraman, founders of 64/1, a curatorial collective that focuses on building a public understanding of how artists and AI can come together to create art for the post-human age (By Madhu Kappurath)

Published: Aug 21, 2018



To create art with artificial intelligence, media and stylistic sensibilities, the artists use a kind of algorithm, called a neural network, and teach it a

<http://www.forbesindia.com/aperture/slideshow/artificial-intelligence-art-parenting-a-new-genre/51059/1#photo>

Forbes India

Headline:

Artificial intelligence art: parenting a new genre

Is Artificial Intelligence the new artist we need to watch out for?

by AVANTIKA BHUYAN
AUGUST 16, 2018

All you need to know about the first exhibit in India to feature art produced by AI

Vogue India

Headline:

Is artificial intelligence the new artist we need to watch out for?



A unique set of works is in the process of being installed at Nature Morte in New Delhi. It includes a set of imaginary landscapes put together by an algorithm, and an animation film based on a short story by Edgar Allan Poe, where each still has been generated by Artificial Intelligence (AI). These form part of Gradient Descent, the first ever AI art show in India.

<https://www.vogue.in/content/is-artificial-intelligence-the-new-artist-we-need-to-watch-out-for/>

Will AI change the art world?

With the first-ever AI art show, the future is here, quicker than we expected.

Chinski Sinha

New Delhi

October 30, 2018 UPDATED: October 30, 2018 19:29 IST



AI artworks

In the AMC series called *Humans*, there is a scene where a homeless man holds a sign that reads "Synth took job" and when Synths (intelligent robots) are awakened to consciousness, humans begin to fear they would unleash their inner artists. And now the last frontiers of human intelligence are being challenged by the legitimisation of art made by Artificial Intelligence as Christie's announces an auction of 11 AI artworks made by the French collective Obvious in October.

Recently, the first-ever AI art exhibition was held in the capital at Nature Morte. Philosophical questions were raised about the ethics and authorship of the art generated by AI. But the co-curators of the show said this is the future and when future comes knocking on the doors, you don't turn away.

From the vantage point of the uninitiated, the frames at the exhibit look like a surrealist collage. Except that these dreams or interpretations were conjured with artificial intelligence working in collaboration with humans. At least that's what the invite for gallery's Gradient Descent said.

While human attributes like memory, language, understanding, reasoning, learning, expression, pattern recognition are being emulated by machines, creativity and spirituality, the two exclusive human privileges, are also now in the threshold zone.



"AI is just interesting," says Peter Nagy, co-director of the gallery. "The holy grail is when the AI starts to have emotions."

But that's not the point of the conversation in this room at the art gallery where the two curators are trying to articulate why AI art must not be dismissed with simplistic arguments like the role of the artist in the making of the art. The point of art is to evoke emotion in the audience, says Karthik Kalyanaraman, the co-curator of the show.

6471 is a curatorial collective based out of Bangalore run by brothers Raghava KK and Kalyanaraman. The collective's aim is to build public understanding on how artists and artificial intelligence can come together to create art for the post-human age. While Raghava KK is a multidisciplinary artist, Kalyanaraman is a former professor of econometrics at University College London and University of Maryland, College Park.

"I think the scope is immense. Young artists I have spoken to are very excited by the medium. However we have to see: it's expensive to train an AI because one needs a GPU (Graphics Processing Unit) and the online services like Amazon's are quite expensive. There is real need for a lab to allow young artists to experiment," Kalyanaraman says.

"A lot will depend on the galleries understanding the new genre here," he adds. According to the gallery, a few artworks from the exhibit have already been sold, which is to say the market is not closed to the idea of such art.

The brothers met Aparajita Jain, the co-director of Nature Morte, some months ago and started the discussion on AI art.

With the show, which got good reviews, the gallery took the risk of aggressively proving the AI art - \$500 to \$40,000.

Among the artworks sold already is a piece done by Wellington artist Tom White, who makes abstracts from AI understandings of everyday items like fans.

While the authorship remains an interesting question, artists feel AI is just another tool for artists to explore their creativity. Kalyanaraman believes AI can change the dynamics of the art world and democratise art.

"Artists have become complacent. When you see art made by AI, you are looking at interpretation," Kalyanaraman says.

With regards to the question of a database of a library of images the technology works with. Nagy says that all art comes from other art.

<https://www.indiatoday.in/lifestyle/what-s-hot/story/will-ai-change-the-art-world-1378860-2018-10-30>

<https://www.indiatoday.in/lifestyle/what-s-hot/story/will-ai-change-the-art-world-1378860-2018-10-30>

India Today

Heading:

Will AI change the art world?

An AI genre in its infancy questions the nature of art

Delhi gallery probes what differentiates the human mark from the machine one

Amy Kazmin



Tom White's AI artworks 'Cello' and 'Electric Fan'. I think of the computer as a tool or — if I'm being gracious — as a collaborator © Tom White

Amy Kazmin in New Delhi

AUGUST 27, 2018

The 20 images hanging at Delhi's Nacore Morie gallery are almost naked, inducing the dark red, meaty texture of human flesh — seemingly the innards of a leg, or a shoulder, or unrecognisable innards. These disquieting images, together entitled "The Anatomy Lesson of Dr Algorithm", were created by artificial intelligence (whether or not they show the influence of Rembrandt, as the name implies, we can leave to the critics). They are one piece of a show entitled Gradient Descent, which consists entirely of original works generated by AI with seven international artists. For Dr Algorithm, Bangalore-based artist Harshit Agarwal, who has a masters degree from MIT's Media Lab, trained his computer on video images of surgical dissections. The AI then generates its own original images of (anatomically incorrect) surgery. "Humans are always thought of in terms of cognition and machines are always thought of in terms of hardware," Mr Agarwal, 26, told a rapt audience at a salon on the eve of the opening. His piece, he says, "is opening up the black box of the human hardware to the machine". With predictions of cognitive computing taking over human jobs, leading to unemployment,

<https://www.ft.com/content/eea358e6-a553-11e8-8ecf-a7ae1beff35b>

THE HINDU

ARTIFICIAL INTELLIGENCE TECHNOLOGY

'Gradient Descent': when Artificial Intelligence meets art

Divya Kala Bhavani

AUGUST 13, 2018 15:30 IST

UPDATED: AUGUST 13, 2018 15:30 IST



'Fall of the House of Usher' by Anna Ridler

On a mission to create a new genre of art, the 'Gradient Descent' show in Delhi observes how artists around the globe are using machines to make masterpieces

Thanks to technology, the roles of the artist and the scientist, as well as their creative processes, are melding and the rigour of the resulting works is

<https://www.thehindu.com/sci-tech/technology/gradient-descent-when-artificial-intelligence-meets-art-641/article24677618.ece>

Financial Times

Headline:

An AI genre in its infancy questions the nature of art

The Hindu

Headline:

The poets of metonymy